

SOFT ATTRIBUTES BENCHMARKING OF [BENCHMARK MODEL NAME]

Vehicle Performance and Benchmarking
April 07th , 2026





Agenda



Introduction

- I. STYLING + II. PERCEIVED QUALITY
- III. HUMAN MACHINE INTERFACE
- IV. ACCOMMODATION & USAGE
- V. THERMAL COMFORT
- VI. VEHICLE SAFETY
- VII. VEHICLE DYNAMICS
- VIII. ENERGY & DRIVABILITY
- IX. NVH
- X. DURABILITY & ENVIRONMENT
- XI. USER ECONOMY
- Conclusion



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Slide 1 - Cover

Include:

- VF vehicle
- Competitor basket
- Market
- VDS Schedule at the date of report release
- Current Program Gate
- Evaluation dates (*KOL, Subjective and Objective Assessment...*)
- Report version
- Data maturity

Suggested title:

VF5 Competitive Benchmark Synthesis



Toyota Tacoma 4th Gen (N400; 2023)

Top 3 Strengths

- **Exceptional reputation for durability & reliability** → proven body-on-frame platform and long-term ownership trust
- **Strong off-road capability across trims** → supported by Crawl Control, Multi-Terrain Select, and TRD hardware
- **Wide product coverage (SR → TRD Pro)** → one model serving fleet, lifestyle, and premium customers

Positioning

- **Segment:** Midsize pickup truck
- **Core Identity:** “Durability-first, capability-driven truck”
- **Brand Positioning:** Benchmark for **resale value, reliability, and off-road credibility**
- **Market Role:** Bridges **daily usability + adventure capability + light-dutywork**

Closing Statement

- Tacoma represents a **highly engineered, modular truck platform** designed to support **multiple lifestyles with one product**
- It reinforces Toyota’s philosophy of **trust, durability, and authentic capability over complexity**
- → A **market benchmark in reliability and off-road performance**, but with opportunity gaps in **electrification and digital differentiation**

Takeaway

- Tacoma = “**Capability + Reliability Benchmark**” in midsize pickup segment
- Future competitors must combine this **DNA with next-gen electrification & smart features**



Why was this car model created? (Product Vision / Market Need)

What does the market need?

What problem does the car solve?

Segment positioning (A/B/C/D, SUV/Sedan/EV/ICE...)

→ Attracts the audience, helping them understand the "reason for existence" of the product.

Đối tượng khách hàng mục tiêu (Target customer)

- Who will buy this car?
- Their behavior/needs?
- Key insights?

Slide 2 - Purpose and decision required

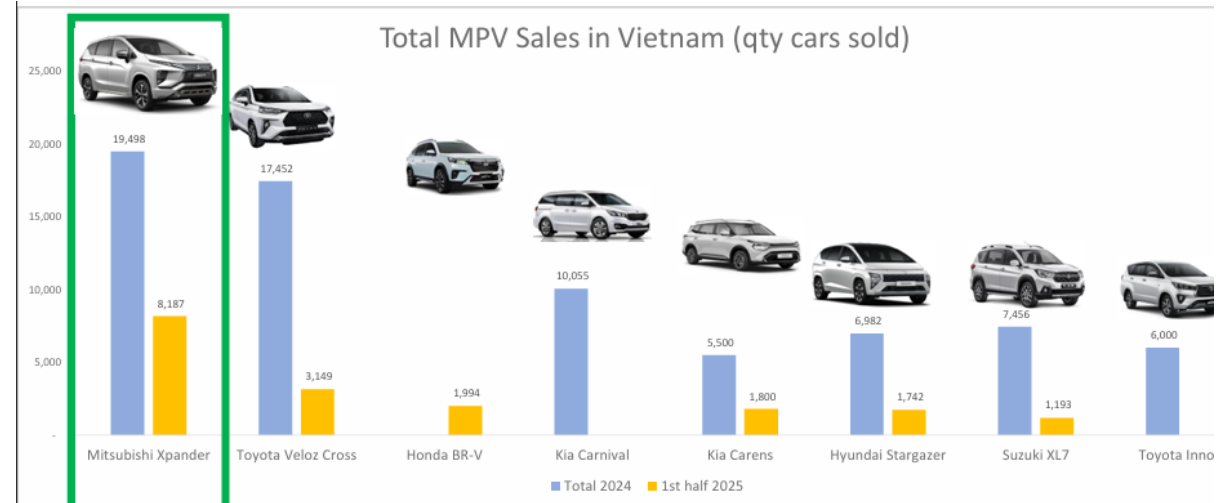
Do not merely state that benchmarking was conducted.

State:

- Why these competitors were selected
- What decision management must make
- Which program milestone the report supports
- Which data streams are included
- Which evidence remains unavailable

The [VCA Benchmarking - Xpander Limo 20251118.pdf](#) provides a useful precedent by explaining why the Mitsubishi Xpander was selected, linking selection to sales performance and Vietnamese customer relevance. However, the report also warns that benchmarking alone is not a definitive conclusion without further market, price, clinic, and strategic analysis.

Why was X-Pander chosen?



Xpander was chosen because, as a **top-selling** model in the MPV segment during its second year, it reflects a strong understanding of **Vietnamese customer preferences**. This success provides us (**Engineering, VCA, PQ, MarCom...**) a foundation for evaluating the key reasons behind its appeal - particularly in terms of appearance.

Benchmark basket and test conditions

Vehicle Identification	VF Vehicle	Competitor 1	Competitor 2
Vehicle / model			
Variant - exact trim			
Model year			
Market specification			
Powertrain type - BEV / HEV / PHEV / ICE			
Seat capacity - occupants			
Production location			

Cars to be assessed

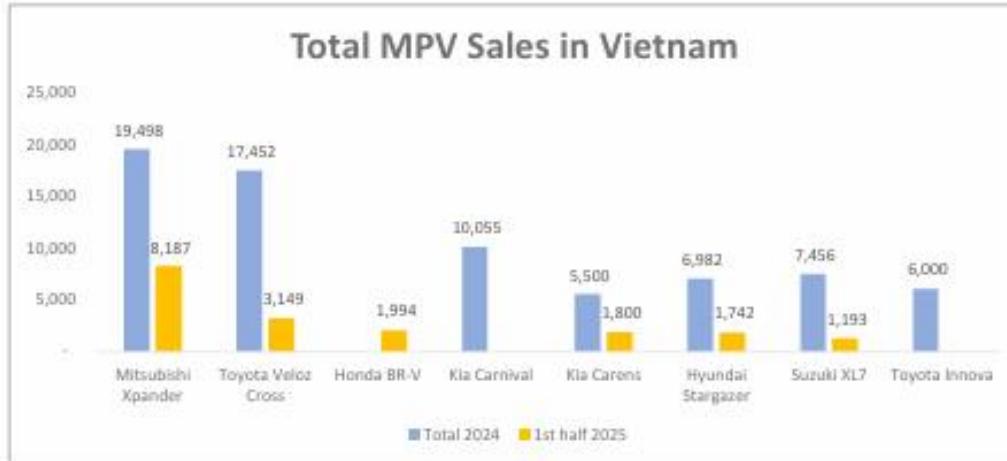


	Xpander	Limo Green
Production Year	2025	2025
Market		
ODO (km)	140,191	39
Driving System	FWD	FWD
Passenger Capacity	7 Seats	7 Seats
Production site	Indonesia	Vietnam
Fuel	Gasoline	EV

Static appearance focus

- Overall Exterior
- Overall Interior
- Finishing
- Visual / Touching
- Customer usage comfortability

Benchmark basket and test conditions



Xpander (Produced in Indonesia)



Variant assessed



AT Premium

Price: 659.000.000 VND



White



AT

Price: 598.000.000 VND



White



MT

Price: 560.000.000 VND



White



New Xpander Cross

Price: 699.000.000 VND



Customers prioritizing comfort and a premium feel for city driving.

Customers seeking versatility and SUV-like aesthetics.

Benchmark basket and test conditions

	Xpander AT Premium	Limo Green
General Information		
Price (VND)	659,000,000 (MT: 560M, AT: 598M, Cross: 699M)	
Dealer network - quantity	67	96
Warranty	3 years / 100,000 km	7 years / 160,000 km
Dimensions - L x W x H, mm	4,595 x 1,750 x 1,750	4,740 x 1,872 x 1,723
Wheelbase - mm	2,775	2,840
Ground clearance - mm	225	170
Curb weight - , kg		
Gross vehicle weight - kg		
Payload - kg		
Luggage capacity - all seats in use, L		
Luggage capacity - rear seats folded, L		
Wheel diameter - mm		
Powertrain		
Maximum power and torque	77 kW and 141 N m	150 kW and 280 N m
Features		
Cruise control	Yes	No - AFS
Entertainment	Apple CarPlay / Android Auto	FM / Bluetooth / USB
Airbags - quantity	6	4
Exterior mirrors	Electric folding and position adjustment	Electric position adjustment - not foldable
Steering-wheel infotainment controls	Yes	No

	Xpander AT Premium	Limo Green
General Information		
Price (VND)	659,000,000 (MT: 560M, AT: 598M, Cross: 699M)	749,000,000
Dealer network (quantity)	67	96
Warranty	3 years/ 100.000 km	7 years/ 160.000 km
Dimension LxWxH (mm)	4595 x 1750 x 1750	4740 x 1872 x 1723
Wheel base (mm)	2775	2840
Ground clearance (mm)	225	170
Powertrain		
Powe Output & Max.Torque	77 kW & 141 Nm	150 kW / 280 Nm
Features		
Cruise control	Yes	No (AFS)
Entertainment	Apple car play/ Android auto	FM/ Bluetooth/ USB
Air bags	6	4
Mirror External	Electric fold/ adjust position	Electric adjust position (not able to fold)
Steering wheel infotainment control	Yes	No

✓ KOL / VOC = Market and customer expectations

Use this slide to answer:

- What is the target customer value?
- What are the segment purchase drivers?
- What causes rejection?
- What is the willingness to pay?
- Which requirements are Must Have, Good to Have, or unnecessary?

For KOL evidence, preserve the voting logic:

- WOW
- MUST HAVE
- GOOD TO HAVE
- ACCEPTABLE
- DO NOT LIKE
- NOT LIKE TO HAVE

The [KOL Feedback for BYD M9 Premium - 260608.pptx](#) already uses this prioritization and records percentages such as 80 percent MUST or 60 percent NOT LIKE TO HAVE. Preserve those results, but map every entry to a PALS code.

VTS data comparison using Color-code (example: Lumin vs. VF3)

Specs for VF – in the Feature List:



Changan Lumin



Highlighted in Red : VF3 inferior
Highlighted in Green : VF3 is better

Parameter	Unit	Trim L
Dimensions	LxWxH (mm)	3,270 x 1,700 x 1,545
Wheelbase	mm	1,980
Curb Weight	kg	925
Max Power	kW	35 / 48
Max Speed	km/h	101
Battery Type	-	LFP
Battery Capacity	kWh	28.08
Range (NEDC)	km	301
Charging Time	0% to 100%	<=10 hrs
Motor Type	-	Permanent Magnet Synchronous
Transmission	-	Fixed Gear
Drive Modes	-	Eco & Sport
Drivetrain	-	FWD

The single available model corresponds to the "L" or "Long Range" spec found in other markets. **Rp 178,000,000 (~\$11,125)**



Slide 7 - Top competitive gaps

Show only the 5 to 8 points that require management attention.
Each line should state:

Customer problem - evidence - VF gap - business impact

Example:
Rear-row storage does not accommodate common bottles and phones - observed by assessors and repeated in KOL feedback - poorer usability than competitor - may reduce family-use satisfaction.
The KOL report identifies insufficient cup holders, phone storage, shoe storage, difficult trunk retrieval, and concerns about second-row ingress. These belong under PALS IV rather than being spread between interior, A&U, and feature chapters. [\[KOL Feedba...m - 260608 | PowerPoint\]](#)

Slide 8 - Top proposals and decision gates

- Possible decisions:
- GO - Implement
 - GO - Validate
 - STUDY
 - HOLD
 - DROP
 - KEEP VF
 - COST-DOWN CANDIDATE

ID	PALS	VF gap	Proposal	KPI	Effort	Owner	Decision

I. STYLING + II. PERCEIVED QUALITY





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Styling + PQ Attributes

- **I.1 Exterior Design**

Design Philosophy (e.g, BYD Ocean Series)



Overall Image

- Large, high-quality images of the vehicle (front, side, rear)

Dimensions & Construction

- Important physical parameters (length x width x height, ground clearance).





Key Exterior Highlights: Design Details, "Wow" Features

- Signature LED Lights
- Main Lines
- Wheels
- Aerodynamics





Styling + PQ Attributes

• I.2 Interior Design

- Images of seats, dashboard, screen, steering wheel
- Materials & Colors
- Legroom/Headroom
- Interior Technology (HUD, central screen, ambient light)

10 INTERIOR ITEMS

- Cabin Overview
- Driver's Position
- Screens & Infotainment
- Control Buttons
- Color, Material, Finish (CMF)
- Seats
- Space & Comfort & Amenities
- Air Conditioning
- Interior Lighting
- Audio
- Smart Technology
- Rear Seats & Trunk





Overall Interior Layout

- Introduction to the cabin overview
 - Design style (minimalist / tech / luxury...)
 - Design philosophy ("human-centered", "driver-focused", "smart cockpit")
- This slide uses a panoramic view of the cabin.





Driver's Cockpit

- Steering Wheel Design (D-cut, 2/3/4-spoke)
- Instrument Cluster (Digital cluster / HUD / TFT)
- Driving Controls (Scroll wheels, mode buttons)
- Ergonomics (Visibility, distance, button layout)



Left Steering Mounted Controls



Right Steering Mounted Controls



Instrument Cluster



Front Row Dashboard



Central Display & Entertainment System

- Screen Size (10", 12.3", 15", etc.)
- Operating System (Android-based, Linux-based, HarmonyOS, etc.)
- Features:
 - Navigation
 - Voice Assistant
 - App Store
 - EV functions (SoC, routing optimization)





Color, Material, Finish (CMF)

Surface materials (soft plastic, leather, faux leather, felt, metal)

- Stitching, patterns, trims
- Interior themes
- → This slide should be zoomed in close-up.





Seating System

- Seat Type: Lumbar Support, Sport, Luxury
- Material: Genuine Leather/Synthetic Leather/Fabric
- Heating/Cooling/Massage
- Front and Rear Space

Front Seats:

- Electric Adjustment
- Heating/Ventilation
- Body-Hugging Design

Rear Seats:

- Shoulder/Leg/Headroom
- Center Armrest, Air Vents
- Folding Capabilities (60:40, Flat Floor...)





Space & Comfort & Amenities

- Legroom, Headroom
- Armrest, Storage Compartments, Charging Ports



Cup Holders - Front



Cup Holders – 2nd row



Cup Holders + 12V Socket – 3rd row



- **Air Conditioning** (Zones, PM2.5 filter, Vent Design)



HVAC Controls on HMI



Front+ Rear Center AC Vents (No 3rd row AC vent)



OMT18-2232

The fragrance is located in auxiliary fascia console center channel

Front outlets



OMT18-2096

Front side outlets



OMT18-2097

Second row outlets



OMT18-2098



Interior Lighting

- Sunroof, Curtains
- Ambient Lighting functions, controls
- Lighting coverage



Sunroof



Front Dome Light

1. Color Of the lamp
2. How many lamps.
3. Is all storage location have lamps.
4. Any gesture control is there to activate or deactivate the lamp?
5. Adjustable Brightness.
6. No blind Spot inside the cabin



Rear Row Roof Mounted Lamps



2nd row Ambient + Reading Light



Hệ thống ánh sáng nội thất (Ambient Lighting)

- Dải LED trang trí
- Số màu (RGB / 64 màu / 256 màu)
- Các chế độ theo cảm xúc hoặc driving mode

Hệ thống âm thanh (Audio System)

- Số loa, thương hiệu (Bose, JBL, Sony...)
- Bộ xử lý âm thanh
- Subwoofer / Surround





Công nghệ & Tiện ích thông minh (Smart Features)

- Sạc không dây
- Cổng USB-C / USB-A
- Ứng dụng kết nối (App Connect, OTA, Smart Key)
- Camera cabin
- Điều khiển giọng nói AI



Charging Ports – Front Row



Charging Ports – Rear Row



Wireless Charger (on dashboard)



Khoang hành lý (Cargo Space)

Gập ghế (60:40 / full-flat) Khả năng mở rộng khi gập ghế

- Dung tích & Không gian đựng đồ
- Tính năng: cốp điện, cảm biến đá cốp



Boot Release Lever/Fuel Lid Release Lever



Open Boot – all rows up



Power Close button



Bootspace Rear Split Seat Folded



Bootspace Rear Split Seat Folded



Boot Compartment



STORAGE BENCHMARK COMPARISON

ITEM	BYD M9	KIA CARNIVAL	DENZA D9	TOYOTA ALPHARD
PIC				
Trunk (7-seat)	~470 L	~540 L	~410 L	~300–350 L
Max Cargo (seats folded)	~2,036 L	~2,785 L	~2,310 L	~1,900 L
Loading Height	Low	Medium	Low	Medium
Tailgate Opening	Wide	Wide	Wide	Narrower
Cabin Storage	High (family-oriented)	High (utility-focused)	High (premium-focused)	Medium
Practical Focus	Balanced family use	Utility & volume	Premium + practicality	Luxury, less cargo-priority





I recommend adding **II.3 Operational Quality** because door-closing sound, switch feel, seat-motor sound, damped movement, latch quality, anti-pinch response, and glovebox movement are repeatedly assessed but do not fit cleanly under visual interior or exterior quality.

For example, the static evaluation records anti-pinch sensitivity, ORVM adjustment noise, seat-motor noise, door-closing sound, and damped mechanisms. These require a permanent home. [\[VF8&9 Stat...D 13042025 | PowerPoint\]](#)

III. HUMAN MACHINE INTERFACE





III. HUMAN MACHINE INTERFACE

- III.1 Infotainment Performance
- III.2 Connectivity
- III.3 HMI Usability
- Controls and Driver Information

Adding Controls and Driver Information avoids forcing instrument-cluster feedback, stalk confirmation, physical HVAC buttons, steering controls, and warning information into general usability.

The [Kia Carnival Subjective Benchmarking.pdf](#) specifically recommends visual stalk-action feedback, a quick-control panel, a customizable home screen, and physical HVAC controls. [\[Kia Carniv...nchmarking | PDF\]](#)



Tacoma – 12.3' Instrument Cluster (Hybrid trims)

[List of Contents](#)



Left Custom Zone (Left Gauge)

- Blank
- Traction Monitor
- Trailer Brake
- Auxiliary Gauges
- Turbo Gauge
- Navigation
- Audio

Center Custom Zone (Main MID Canvas)

- Blank
- Settings
- Messages
- Driving Support

Right Custom Zone (Right Gauge)

- Blank
- Electric Boost Gauge
- Fuel Economy
- Driving Support
- Navigation
- Audio
- Drive Info
- Trip A
- Trip B
- Tire Pressure
- Pitch and Roll
- Trailer Mode
- Blind Spot Monitoring
- Engine / Transmission Oil Temperature

Software Restrictions and Layout Flaws

Asymmetrical Menu Locking

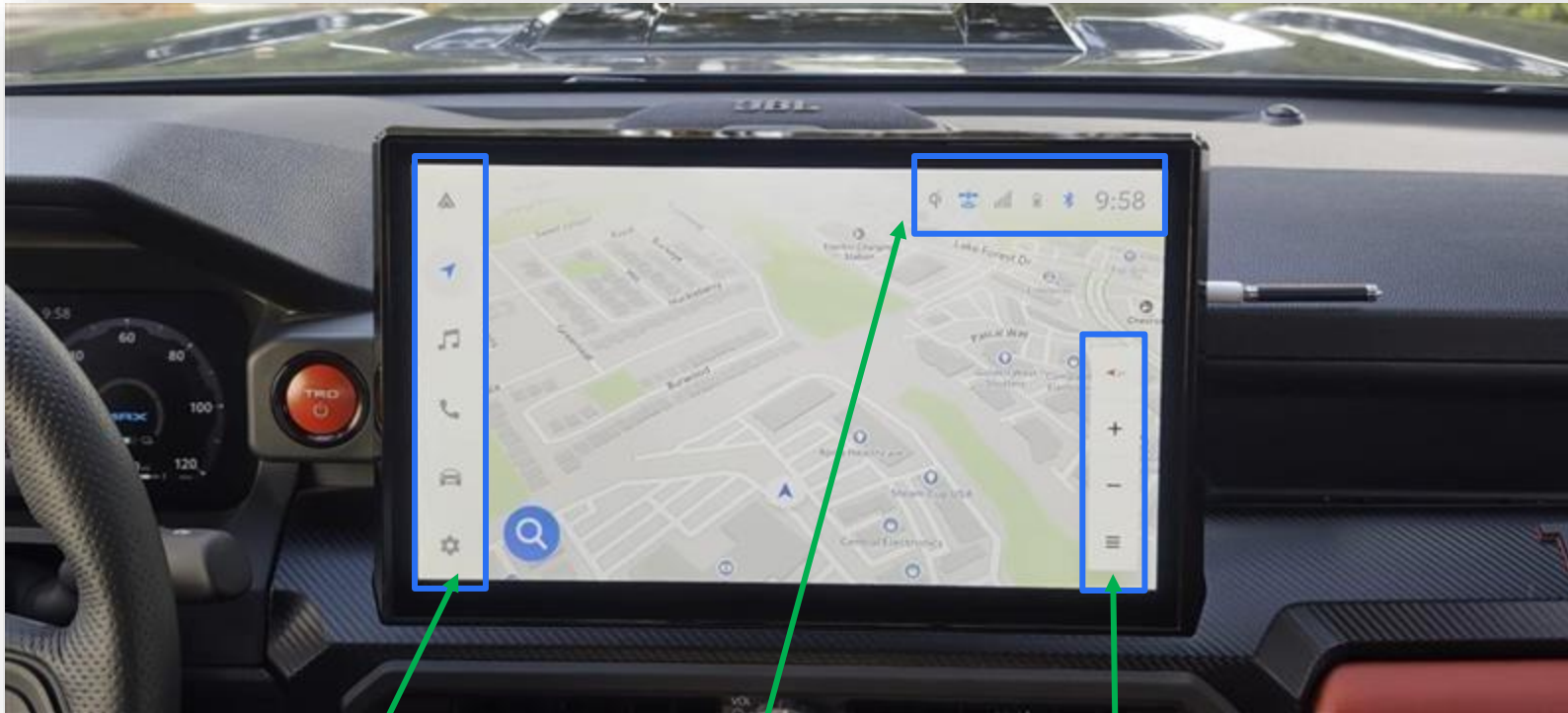
- Restricts identical data cards from being assigned to both outer gauge zones simultaneously.
- Prevents critical tracking features—such as the electric boost gauge or fuel economy history—from being mapped onto the left dial, forcing them exclusively onto the right side.
- Forces the driver to cycle through menus repeatedly rather than allowing a side-by-side view of favorite metrics like boost pressure and fuel economy at the same time.

Center Canvas Under-Utilization

- Severely limits data configuration choices within the primary center multi-information display sector.
- Restricts options almost entirely to a completely blank screen or the active driver safety assistance layout.
- Misses an opportunity to use the largest portion of the screen canvas for deeper, custom driver parameters or alternative maps.

Hybrid Energy Telemetry Omission

- Deletes standard real-time hybrid powertrain energy flow schematics from the digital dashboard instrumentation.
- Fails to provide visible regeneration levels or instant battery-to-motor power transfer graphics to the driver.
- Abandons visual hybrid tracking tools that have been standard across legacy Toyota architectures since the second-generation Prius.



Primary Navigation Rail (Left Persistent Menu)

- Phone Projection
- Navigation
- Media
- Phone
- Vehicle Settings
- General Settings

Header Display Constants

- Digital Clock
- Connected Device Status
- Cellular Signal Strength Indicator

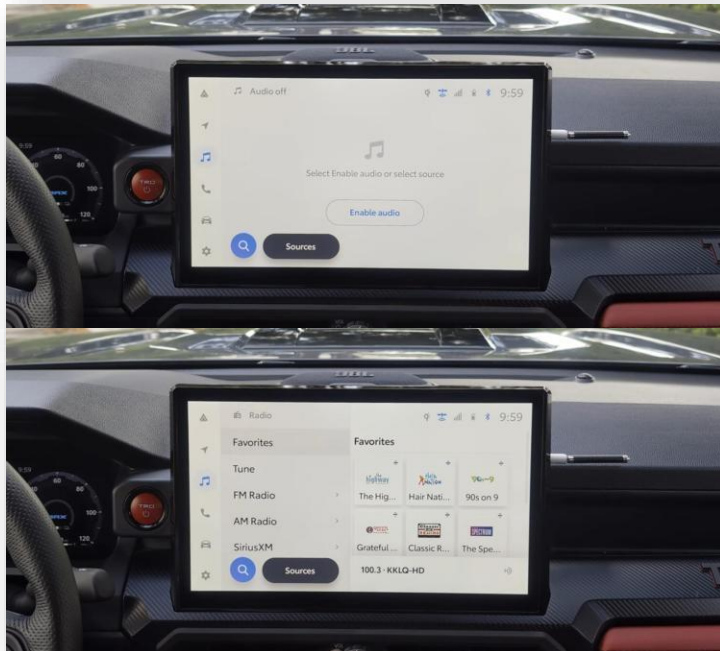
Navigation Module Interface

- Google Maps Base View
- Manual Zoom Controls (+ / - Buttons)
- Pinch-and-Zoom Gesture Interface
- Orientation Control (North Up Toggle)



Tacoma – 14' HMI

List of Contents



Media Hub Source Menu

- Audio Enable Toggle
 - Favorite Channels Strip
 - Manual Tune Keyboard
 - AM Radio
 - FM Radio
 - HD Radio (NPR Integration)
 - SiriusXM Satellite Radio

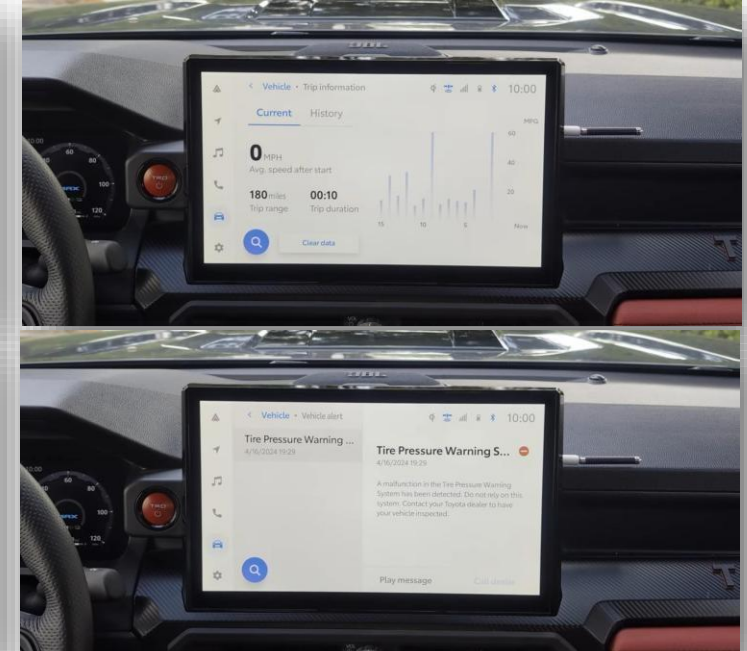
Phone Communication Sub-Menu

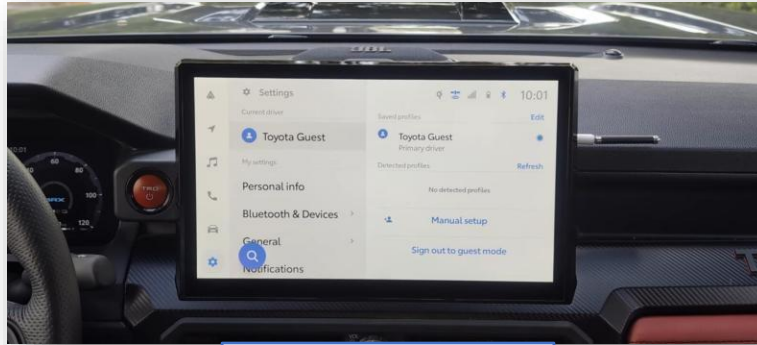
- Dial / Call Panel
- Message Log View
- Favorite Contacts List



Vehicle Telemetry Screen

- Trip Information
 - 15-Mile Average Fuel Economy Bar Chart
 - Trip Range Estimate
 - Trip Duration Counter
 - History (Day-to-Day Log)
- Vehicle Alerts
 - Malfunction Warning Monitor (ex: Tire Pressure System Log)





Settings Main Page

Settings Tree Architecture

Profile Management

- Cloud User Profiles
- Media Preferences Sync
- Climate Preferences Sync

Personal Info

- Bluetooth Devices and Keys
- Driver Identification Data
- Manage Devices Menu (Add Another Device / Search / Remove Device)

General Settings

- Accessibility (Screen Beep On/Off Toggle)
- Screen Touch Sensitivity Adjustment
- Date and Time (Set by GPS Toggle / Manual Augment / 24-Hour Format)
- Keyboard Layout Menu
- Language and Units (English / Spanish / French Selection)
- Measurement Metric Toggle (Kilometers / Miles Selector)
- Notifications Filter Settings
- Wi-Fi Sub-Tree
- Wi-Fi Hotspot (Monthly Paid Cellular Data)
- Wi-Fi Receiver (Home Network Over-the-Air Update Link)



Dark Appearance on HMI

- ⑩ Display Settings
- ⑩ Screen Brightness Slider
- ⑩ Screen Contrast Slider
- ⑩ Display Off Toggle
- ⑩ Daytime / Nighttime / Automatic Theme Toggle (Dark Mode)
- ⑩ Sound and Media
- ⑩ Audio Equalizer (Treble / Mid-range / Bass Sliders)
- ⑩ Phone Ringtone Volume Slider
- ⑩ System Voice Volume Slider
- ⑩ Navigation Setup Sub-Tree
- ⑩ Voice and Search (Wake Word "Hey Toyota" Activation Toggle)
- ⑩ Vehicle Customize
- ⑩ Light Settings (Exterior Light-Off Delay Timer)
- ⑩ Daytime Running Lamps (DRL On/Off Toggle)

IV. ACCOMMODATION & USAGE





IV. ACCOMMODATION & USAGE

- IV.1 Occupant Posture and Primary Controls
- IV.2 Roominess
- IV.3 Visibility
- IV.4 Ingress and Egress
- IV.5 Reach and Accessibility
- IV.6 Cargo and Stowage
- Seating Flexibility and Comfort

Add Seating Flexibility and Comfort

because seat folding, captain-seat functions, third-row recline, massage, ventilation, seat travel, and seat mechanisms form a major MPV decision area.

The KOL report treats third-row folding, captain-chair headrests, second-row massage, and electric third-row adjustment as important recommendations, while the static report separately evaluates seat travel, third-row access, thigh support, and folding arrangements. [\[KOL Feedba...m - 260608 | PowerPoint\]](#),

[\[VF8&9 Stat...D_13042025 | PowerPoint\]](#)

V. THERMAL COMFORT





V. THERMAL COMFORT

- V.1 AC Performance
- V.2 Heating Performance
- Air Quality and Thermal Features

Required:

- Air Vent locations
- Quantity of vents

Add Air Quality and Thermal Features for:

- PM2.5 filtration
- Air purifier
- AQI display
- Solar glazing
- Sunshades
- Seat ventilation and heating
- Thermal preconditioning



HVAC Controls on HMI



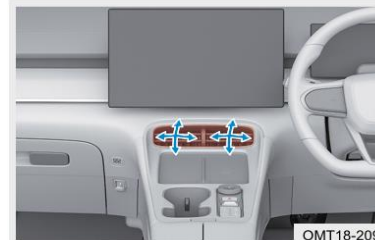
Front+ Rear Center AC Vents (No 3rd row AC vent)



OMT18-2232

The fragrance is located in auxiliary fascia console center channel

Front outlets



OMT18-2096

Front side outlets



OMT18-2097

Second row outlets



OMT18-2098



HVAC/THERMAL SUBJECTIVE ASSESSMENT

HVAC Assessment
among benchmarks

Criteria	BYD M9 (DM-i PHEV)	Denza D9 (PHEV)	Kia Carnival 2.2D Premium
Cabin Cooling Performance (Hot Climate)	Strong cooling performance expected; large glass area MPV requires high-capacity compressor. Comfort-oriented airflow tuning.	Premium-class cooling performance; emphasis on rear passenger comfort and airflow distribution.	Effective cooling; conventional ICE-driven A/C compressor with stable performance in hot climates.
Rear Cabin HVAC Support	Multi-zone climate with dedicated rear vents; designed for 2nd/3rd row comfort.	Advanced rear climate control focus; premium airflow refinement for executive MPV positioning.	Rear vents available; airflow adequate but less refined distribution compared to premium MPV.
Compressor Type	Electrically driven compressor (typical PHEV architecture; allows A/C operation in EV mode).	Electrically driven compressor (PHEV architecture).	Belt-driven compressor (engine-dependent).
Low-Speed / Idle Cooling	Stable cooling in EV mode without ICE engagement.	Stable EV-mode cooling; smooth transition when ICE activates.	Cooling performance depends on engine speed; slight variation at idle possible.
Battery Thermal Management	Integrated liquid cooling for traction battery; thermal control critical for PHEV efficiency.	Advanced battery thermal management aligned with premium positioning.	No HV battery; conventional engine cooling system.
Cabin Heating Strategy	Likely electric heater support for EV operation (verify heat pump availability).	Premium heating performance; possible advanced thermal management (verify heat pump).	Diesel engine waste-heat based heating; strong heating in cold conditions once engine warms.
Thermal Comfort Consistency (Full Load 6–7 pax)	Comfort-focused MPV tuning; performance dependent on airflow calibration and glass insulation.	Strong thermal comfort consistency expected in full occupancy.	Adequate for family use; may require higher fan speed in extreme heat.

Note: Assessment synthesized from publicly available specifications, system architecture characteristics, and media road-test impressions..

VI. VEHICLE SAFETY





VI. VEHICLE SAFETY

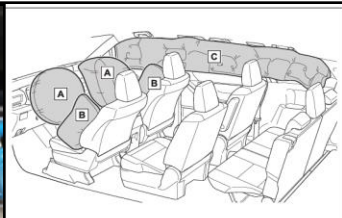
- VI.1 Safety Standards
- VI.2 ADAS
- VI.3 Passive and Active Safety Features
- Functional Safety and Misuse Protection

Functional Safety and Misuse Protection

should capture anti-pinch behavior, emergency releases, charging-door safety, child safety, warning logic, and visibility-related operation.



INNOVA HYCROSS
SCORES 5-STAR
IN BHARAT NCAP



6 SRS Airbags

Summary of Scores

14.47 out of 16.00
Frontal Offset Deformable Barrier Test

16.00 out of 16.00
Side Moveable Deformable Barrier Test

Test Results

● Good ● Adequate ● Marginal ● Weak ● Poor

Frontal offset deformable barrier test

Side moveable deformable barrier test

Side pole impact test (Pole) : OK

Bharat NCAP 5-star

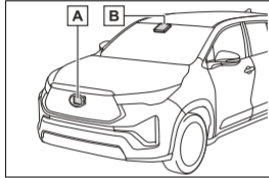
6 SRS Airbags

Sensors used by Toyota Safety Sense

Two types of sensors are used to obtain the necessary information for system operation.

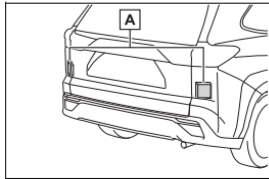
■ Sensors which detect the surrounding conditions

► Front



A Front radar sensor
B Front camera

► Rear



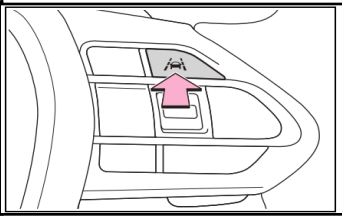
A Rear side radar sensors



Front Camera (standard feature)



Rear Camera (optional)



PCS version	Detectable objects	Function availability	Countries/areas
1	• Vehicles • Bicycles* • Pedestrians • Motorcycles* • Walls	The pre-collision warning, pre-collision brake assist, pre-collision braking, emergency steering assist, intersection collision avoidance support, low speed sudden acceleration suppression function	Brunei
2	• Vehicles • Bicycles* • Pedestrians • Motorcycles*	The pre-collision warning, pre-collision brake assist, pre-collision braking	Indonesia, Cambodia, Thailand, Philippines, Vietnam, Malaysia, Laos, Guatemala
3	• Vehicles	The pre-collision warning, pre-collision brake assist, pre-collision braking, suspension control	India

*: Detected as a detectable object only when being ridden.

The countries and areas for each region listed in the table are current as of March 2025. However, depending on when the vehicle was sold, the countries and areas of each region may be different. Contact your Toyota dealer for details.

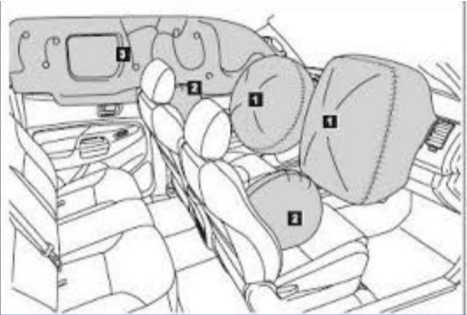
Lane Tracing Assist (LTA) switch on the steering wheel



Safety Attribute

[List of Contents](#)

Crash-test ratings	Passive safety	Vehicle Safety Protection System	ADAS features	Adaptive Cruise Control feedback - features	Structural Safety Norms
● NHTSA: 3★ frontal / 5★ side • IIHS: Good ratings	8 airbags (front, side, curtain, knee)	Toyota Safety Sense 3.0	PCS, DRCC, LTA, RSA, PDA, AHB	Full-speed radar cruise with stop & go	High-strength steel reinforced cabin



2025 TOYOTA TACOMA

PU/CC 4WD



VEHICLES

4/5 Overall Safety Rating



Recommended Safety Technologies

Meets Performance Criteria 4

[Detailed Results >](#)

Recalls

4

Investigations

0

Complaints

55

Airbag locations in the cabin



Ramp Travel Index:

- 510 (Base trims)
- Trailhunter: 532
- TRD Pro: 520

[2025 TOYOTA TACOMA | NHTSA](#)

VII. VEHICLE DYNAMICS





VII. VEHICLE DYNAMICS PIC: Prabakar	
VII.1	RIDE
VII.2	HANDLING
VII.3	BRAKING
VII.4	PERFORMANCE
VII.4	STEERING

Criteria	BYD M9 (DM-i PHEV)	Denza D9 (PHEV)	Kia Carnival 2.2D Premium
Ride Comfort – Urban	Compliant and comfort-oriented; EV operation enhances smoothness in city driving.	Very comfortable, premium ride quality with refined suspension tuning.	Comfortable and family-oriented; absorbs typical city bumps effectively.
Ride Comfort – Highway	Stable at cruising speed; tuned more for comfort than dynamic firmness.	Feels planted and refined at high speed; strong composure for large MPV.	Stable but slightly soft damping character at higher speeds.
Body Control (Roll / Pitch)	Moderate body roll typical of comfort-focused MPV with higher center of gravity.	Better body control than mainstream MPVs; still comfort-biased rather than sporty.	Noticeable roll under aggressive cornering; comfort prioritised.
Steering Feel	Light and easy; limited feedback; not performance-focused.	Smooth and progressive; not sporty but more confidence-inspiring.	Light steering, tuned for ease of use rather than feedback.
Handling Agility	Large mass noticeable in dynamic maneuvers; predictable but not agile.	More composed than typical MPVs; balanced for premium comfort.	Predictable and safe; agility limited by size and diesel powertrain character.
Powertrain Response	Strong initial EV torque; ICE engagement may slightly alter response under load.	Smooth torque delivery; refined blending between electric and ICE modes.	Strong low-RPM diesel torque; less immediate response compared to EV/hybrid launch.

Note: Assessment synthesized from multiple international media first-drive reviews and road tests, including regional automotive publications and user driving impressions.

VIII. ENERGY & DRIVABILITY





What to include:

VIII. ENERGY & DRIVABILITY PIC: Kulasekharan	
VIII.1	RANGE
VIII.2	DRIVABILITY
VIII.2	CHARGING TIME



EMDQ Fuel Consumption

[What is the 2024 Ford Ranger Gas Mileage?](#)

[List of Contents](#)

[Xe bán tải Ford Ranger có bình dầu bao nhiêu lít?](#)

Trim and Powertrain Configuration	City Fuel Economy	Highway Fuel Economy	Combined Fuel Economy
USA: Ranger XLT (4x2) 2.3L EcoBoost Gasoline (270 hp)	11.2 L/100km (21.0 MPG)	9.4 L/100km (25.0 MPG)	10.7 L/100km (22.0 MPG)
USA: Ranger Raptor (4x4) 3.0L EcoBoost V6 Gasoline (405 hp)	14.7 L/100km (16.0 MPG)	13.1 L/100km (18.0 MPG)	13.8 L/100km (17.0 MPG)
VIETNAM: Ranger XLS (4x2 AT) 2.0L Single-Turbo Diesel (170 hp)	8.5 L/100km (27.7 MPG)	7.0 L/100km (33.6 MPG)	8.0 L/100km (29.4 MPG)
VIETNAM: Ranger Raptor (4x4) 2.0L Bi-Turbo Diesel (207 hp)	9.5 L/100km (24.8 MPG)	6.77 L/100km (34.7 MPG)	7.78 L/100km (30.2 MPG)
VIETNAM: Ranger Raptor (4x4) 3.0L EcoBoost V6 Gasoline (392 hp)	14.9 L/100km (15.8 MPG)	10.1 L/100km (23.3 MPG)	11.5 L/100km (20.5 MPG)



EMDQ, Range with Full Tank

Trim and Powertrain Configuration	Full Tank Capacity	City Driving Range	Highway Driving Range	Combined Driving Range
USA: Ranger XLT (4x2) 2.3L EcoBoost Gasoline (270 hp)	<u>18.76 Gallons</u> <u>(71 Liters)</u>	~635.4 km	~756.4 km	~665.6 km
USA: Ranger Raptor (4x4) 3.0L EcoBoost V6 Gasoline (405 hp)	<u>20.34 Gallons</u> <u>(77 Liters)</u>	~522.7 km	~588.0 km	~555.4 km
VIETNAM: Ranger XLS (4x2 AT) 2.0L Single-Turbo Diesel (170 hp)	<u>85.8 L</u>	~941.2 km	~1,142.9 km	~1,000.0 km
VIETNAM: Ranger Raptor (4x4) 2.0L Bi-Turbo Diesel (207 hp)	<u>80.0 L</u>	~860.2 km	~1,269.8 km	~1,028.3 km
VIETNAM: Ranger Raptor (4x4) 3.0L EcoBoost V6 Gasoline (392 hp)	<u>80.0 L</u>	~536.9 km	~792.1 km	~695.7 km



Towing & Payload

[List of Contents](#)

Trim and Powertrain Configuration	Maximum Braked Towing Capacity	Maximum Payload Capacity	Primary Rear Suspension Layout and Towing Hardware
USA: Ranger XLT (4x2) 2.3L EcoBoost Gasoline (270 hp)	3,401 kg (7,500 lbs)	818 kg (1,805 lbs)	Traditional multi-leaf springs with outboard-mounted dampers
USA: Ranger Raptor (4x4) 3.0L EcoBoost V6 Gasoline (405 hp)	2,500 kg (5,510 lbs)	640 kg (1,411 lbs)	Performance-focused Watts-linkage with coil springs and FOX Live Valve shocks
VIETNAM: Ranger XLS (4x2 AT) 2.0L Single-Turbo Diesel (170 hp)	3,500 kg (7,716 lbs)	703 kg (1,550 lbs)	Heavy-duty commercial multi-leaf suspension optimized for maximum cargo weight
VIETNAM: Ranger Raptor (4x4) 2.0L Bi-Turbo Diesel (207 hp)	2,500 kg (5,512 lbs)	409 kg (902 lbs)	Off-road high-travel suspension with a Watts-link system and aluminum control arms
VIETNAM: Ranger Raptor (4x4) 3.0L EcoBoost V6 Gasoline (392 hp)	2,500 kg (5,512 lbs)	409 kg (902 lbs)	Off-road high-travel suspension with a Watts-link system and FOX 2.5 Live Valve units



EMDQ Battery / Charging

Battery management system supplier	
High-voltage Battery Size (Gross/Net)	
High-voltage Battery Weight	
High-voltage Battery Chemistry	
Pure EV Range NEDC, km	
Pure EV Range WLTP, km	
Consumption (kWh/100km)	
Battery Cooling Options	
Regulations for High-Voltage Systems (if applicable)	
Battery degradation rates under typical climates	

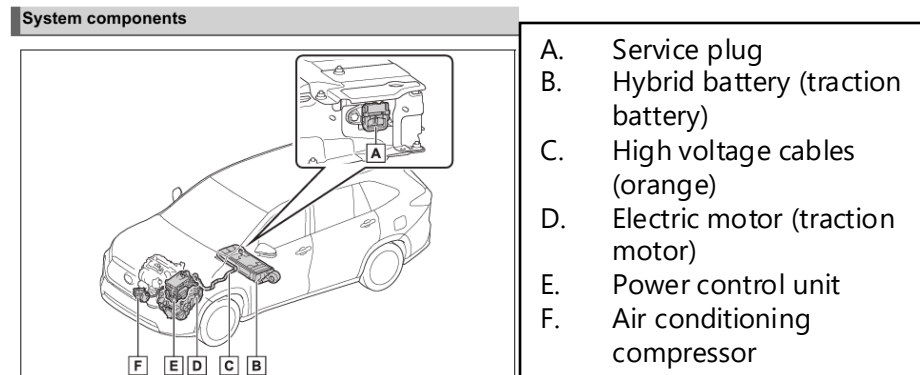


air intake vent under the **left front seat** (independent of whether **LHD or RHD**) = cooling the hybrid battery



EMDQ, EV Powertrain

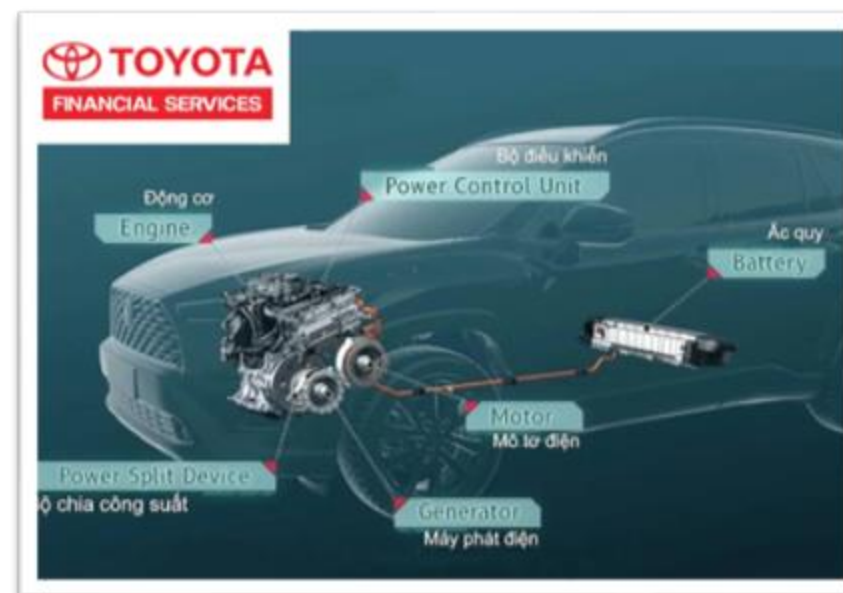
Electric Motor Type	
Electric Motor Power & Torque	
Inverter type and switching frequency	
Regenerative braking strategies: levels, maximum recuperation power, one-pedal mode availability	
Vehicle control unit capabilities: torque vectoring logic, traction control sensitivity.	
Top Speed, km/h	
Acceleration 0-100km/h	
Riding Modes	





EMDQ ICE Powertrain

ICE Engine	Power & Torque	Compression ratio	ICE Fuel Type	Environmental and Emissions Certs	Thermal management architecture / Cooling performance	Fuel injection type	Fuel Tank Capacity, liters	Combined Range, km	Fuel Consumption, L/100km





Transmission

Transmission type / Available modes	Gear ratios, spread, and shift logic	Thermal load capacity of the transmission	Hill-hold, creep tuning, low-speed refinement



e-CVT on Innova HEV



Chassis & Performance

- Suspension (MacPherson, Multi-link...)
- Steering
- NVH
- Driving Modes
- → An illustration of the chassis layout should be included.



Front: MacPherson Strut Independent



Rear: Multi-Link Independent



Drive Mode Buttons/Terrain Selector



Bottom View: Fuel Tank (with anti-gravel coating)



Fuel Tank Is The Bottom Point Of The Vehicle
(140mm Clearance)



Weight Attribute (source: a2mac1)

Weight (Curb) + Axle Distribution	
Weight (Gross) + Axle Distribution	
Center of Gravity at Curb	
Payload of the vehicle	
High-voltage Battery Size (Gross/Net)	
High-voltage Battery Weight	
High-voltage Battery Chemistry	

Link app:

[https://ibp.a2mac1.com/data/product/A0000065YAP2EU01?productTypeGroupId=\\$0000000000000001](https://ibp.a2mac1.com/data/product/A0000065YAP2EU01?productTypeGroupId=$0000000000000001)

TK: v.vinhhpt2@vinfast.vn

MK: Hptv2003#####

IX. NVH





IX. NVH

- IX.1 Powertrain NVH and Sound Quality
- IX.2 Road NVH
- IX.3 Wind Noise
- IX.4 Climate-Control NVH
- IX.5 Operational Sound Quality
- IX.6 Buzz, Squeak and Rattle

Add IX.5 and IX.6. The Kia Carnival_Subjective Benchmarking.pdf</File separately identifies HVAC noise, wiper noise, door operation, tyre noise, wind noise, BSR, tactile vibration, and sealing. Combining all of them into four headings would lose diagnostic clarity. [\[Kia Carniv...nchmarking | PDF\]](#)



NVH benchmarking - Overall Scope

Category	Description
Objective	Evaluate and compare Noise, Vibration, and Harshness performance and solutions of benchmark vehicles
Benchmarking Type	Soft benchmarking (internet, teardown visuals, forums, user reviews) and deep dive where A2Mac1 data is available
Focus Areas	Acoustic packaging, noise-generating component isolation, tactile vibration, perceived noise performance
Key Outputs	Feature comparison, NVH performance indications, user-reported issues, strengths and weaknesses

Checklist – What topics to search for

Section	Required Content
Acoustic packaging	Material and feature list
Component NVH	Packaging and isolation analysis
Performance NVH	Idle, part throttle, WOT, high speed
Tactile vibration	Steering, pedals, seat
User issues	Complaints and feedback
Summary	Strengths, weaknesses, observations
References	All data sources used



Acoustic Packaging & Insulation Materials

Subsystem	Items to Benchmark	What to Check
Firewall	Insulation pads, thickness, coverage	Presence, visual quality, perceived effectiveness
Hood	Hood insulator	Coverage area, material type
Floor	Floor insulation, carpet backing	Extent of coverage, damping presence
Doors	Door acoustic pads, trim layers	Single vs multi-layer construction
Trunk	Trunk insulation, spare wheel well treatment	Noise isolation quality
Roof / Ceiling	Headliner absorption material	Relation to rain and wind noise
Glass	Acoustic windshield, acoustic side glass	Front only or front + rear

Damping Solutions Benchmarking

Area	Damping Elements	Evaluation Criteria
Floor panels	Damping sheets / pads	Coverage level (low / medium / high)
Body panels	Sheet metal damping	Strategic placement or minimal usage
Engine bay	Structural damping patches	Visibility and intent



NVX Component Acoustic Packaging

Component	Packaging Aspect	Benchmark Focus
Compressor	Acoustic cover, mounting isolation	Encapsulation presence and quality
Radiator fan motor	Housing, isolation	Noise containment
Rotating components	Decoupling, mounts	Vibration transmission risk
Motors / Pumps	Shielding, covers	Structural and airborne noise control
Mounting systems	Rubber / hydraulic mounts	Vibration isolation design

Active Noise Control (If Applicable)

Feature	Description	Notes
Active Noise Cancellation	Cabin noise cancellation using microphones and speakers	Presence / absence
System Integration	Microphone and speaker layout	Publicly available information
Target Noise	Powertrain / road noise reduction	Claimed effectiveness

Idle Condition NVH Performance

Condition	Evaluation Criteria
Idle – AC OFF	Cabin noise, steering vibration
Idle – AC ON	Noise increase, vibration change
Steering wheel	Tactile vibration perception
Pedals	Accelerator and brake vibration
Seat	Low-frequency vibration presence

Part Throttle / Light Acceleration NVH

Driving Scenario	NVH Parameters
City driving	Noise smoothness, vibration level
Highway cruising 40–60 km/h	Steering vibration, cabin noise
Highway acceleration 60–80 km/h	Noise character, booming/drone

Wide Open Throttle (WOT) NVH

Condition	Benchmark Focus
Full throttle acceleration	Steering wheel vibration severity
Pedal feedback	Accelerator and brake vibration
Noise intrusion	Engine or drive unit noise
Harshness	Sudden NVH degradation

High-Speed NVH & Wind Noise

Area	Evaluation Points
Doors	Wind noise leakage
Windshield	Airflow noise
Door mirrors	Whistle or turbulence noise
Cabin	Noise stability at high speed



Tactile Vibration Assessment

Contact Point	Description
Steering wheel	Primary tactile NVH indicator
Accelerator pedal	Vibration during acceleration
Brake pedal	Vibration during braking
Seat	Low-frequency body vibration



Data Sources for NVH Benchmarking

Source	Usage
OEM websites	Feature availability
Press kits	Claimed NVH technologies
YouTube	Visual and subjective NVH checks
Owner forums	Real-world NVH issues
Reviews	Comparative NVH impressions
A2Mac1	Detailed materials and architecture (if available)

NVH SUBJECTIVE ASSESSMENT

NVH Criteria	BYD M9 (DM-i 5.0 PHEV)	Denza D9 (PHEV)	Kia Carnival 2.2D Premium (Diesel)
Urban – Background Quietness	Very quiet in EV mode; ICE engages depending on SOC/load.	Quiet in EV mode; ICE engagement reported smooth and refined.	Noticeably quieter than previous gen, but diesel idle still perceptible.
Highway – Wind Noise	Wind noise reported controlled; not positioned as full luxury benchmark.	Highway composure praised; premium sealing and glazing noted.	Stable at cruising speeds; wind noise acceptable but tire/road noise increases on coarse asphalt.
Coarse Asphalt – Road/Tire Noise	Road noise dependent on tire spec; MPV weight may expose low-frequency booming.	Reported well insulated from road imperfections; premium NVH tuning focus.	Diesel variant shows more noticeable road/tire noise on rough surfaces.
Kickdown / Hill Climb	ICE ramp audible under heavy load; typical PHEV behavior.	ICE engagement described as smoother than mainstream hybrids.	Diesel engine clearly audible under acceleration/load.
Powertrain Tonality	Possible ICE transition + e-motor tonal character; no major complaints reported.	Refined power delivery; minimal harshness reported.	Diesel combustion noise more pronounced at idle and hard throttle.
BSR (Rattle / Squeak)	No systematic data available; requires physical validation.	Build quality praised in reviews.	Mixed owner feedback; not flagged as major systemic issue.

Note: NVH comparison is based on publicly available media reviews and owner feedback. Source: AutoHome, Dongchedi, Paultan, OneShift, Car and Driver, MotorTrend, US & SEA

X. DURABILITY & ENVIRONMENT





X. DURABILITY & ENVIRONMENT		
X.1	STRENGTH	-VTS Target: AFS global warranty Considering : 10 years / 200.000 km for warranty -Competitors :Commercial 1. Goldan BM Reference vehicle :Toyota Tacoma :3 Years / 160,000 km/ 7 Years 2.Silver- Ford Maverick: 160.00 years 3.Technical BM- Ford Ranger: 100.000 km / 5 years -VF Key :Internal Benchmark 1. VF Wild 200.000 KM/ 10 Year (VN-Market) 2. VF8 and VF9 Legacy 200.000 KM/ 10 Year (EU/US)
X.2	CORROSSION	-VTS target proposal (Corrosion perforation): 12 years (AFS Warranty NA) -Attribute Brand DNA (C-Segment): 7 ~ 12 years -Competitors Warranty Policy (Corrosion perforation): 1. Goldan BM Reference vehicle :Toyota Tacoma :5 Years unlimited mileage 2.Silver- Ford Maverick: 5 years unlimited mileage 3.Technical BM- Ford Ranger: 5 Years unlimited mileage -VF Key :Internal Benchmark 1. VF Wild : 10 Year (AFS VN-Market) 2. VF8 and VF9 Legacy :12 Year (EU/US)
X.3	TIGHTNESS	-VTS target proposal (Water Wading, maximum depth): 500 mm (228 mm ground clearance) (assumption, same as BEV) -Attribute Brand DNA (C-Segment): 450 ~ 600 mm -Competitors Warranty Policy (Corrosion perforation): 1. Goldan BM Reference vehicle :Toyota Tacoma :762mm 2.Silver- Ford Maverick: Not published (est ~450mm) 206mm Ground clearance 3.Technical BM- Ford Ranger: 800 - 850mm -VF Key :Internal Benchmark 1. VF Wild VN : 500mm



DURABILITY SUBJECTIVE ASSESSMENT

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Durability Criterion		
Buzz, Squeak, and Rattle over time		
Chassis and Suspension Degradation		
Underbody and Frame Rust Resistance		
Surface and Cosmetic Corrosion		
Cabin Sealing and Water Wading Tightness		

Note: Assessment synthesized from multiple international media first-drive reviews and road tests, including regional automotive publications and user driving impressions.

XI. USER ECONOMY





XI. USER ECONOMY PIC: Rajesh (not applicable with Asia market)	
XI.1	BUYING COST
XI.2	SERVICE & MAINTENANCE

XI. USER ECONOMY

- Purchase Cost
- Energy Cost
- Service and Maintenance
- Warranty and Ownership Assurance
- Residual Value
- Insurance and Repairability

This chapter should represent total cost of ownership, not only purchase and service costs.

Recalls / Service Issues





Potential Loss of Brake Performance - NHTSA Recall No. 25V-058

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TOYOTA

SAFETY RECALL 25TA04 *(Remedy Notice)*

Certain 2024 - 2025 Model Year Tacoma 4x4 Vehicles with Toyota 17" and 18" Wheels
Certain 2024 - 2025 Model Year Tacoma 2wd Vehicles with Toyota 17" and 18" Wheels
Potential Loss of Brake Performance
NHTSA Recall No. 25V-058



STOP! DO NOT SELL NEW VEHICLES IN DEALER INVENTORY.

[Refer to the Dealer Inventory Procedures section for more details.](#)



Rear Brake Hose Damage (NHTSA Campaign: 25V-058): Mud and dirt buildup inside the wheels during off-road use can wear down the rear brake hoses. This can cause a brake fluid leak, decreasing stopping performance



Variants & Colors at a specific market (example: Indian Market Toyota Innova Hycross HEV)

In India, the **Toyota Innova Hycross** is available in 10-12 variants across seven primary trims, split between gasoline and strong-hybrid powertrains.

Available Trims

The lineup is divided by engine type, with all models featuring an automatic transmission (CVT/e-CVT).

- **ZX (O):** The top-tier hybrid trim that includes the [Toyota Safety Sense \(ADAS\)](#) suite with adaptive cruise control and lane trace assist.

Engine Type	Trims (7/8 Seater Options)	Key Features
Gasoline (2.0L Petrol)	G-SLF, GX, GX (O)	16-inch alloy wheels, 8-inch or 10.1-inch touchscreen, and basic safety tech.
Strong Hybrid (2.0L + Electric)	VX, VX (O), ZX, ZX (O)	Panoramic sunroof, 18-inch wheels, ventilated seats, and 9-speaker JBL system.

Dealership Cost Structure at the SOP in the Indian Market (end-2022)

NANDI TOYOTA												TOYOTA	
Price Effective from 01-Jan-2023													
Model	Innova Hycross	Suffix	Ex Showroom	Tax Collected at Source*	Road Tax Cost	Registration Cost**	Accessories Basic Kit Cost***	Insurance Cost [1 Year, Zero Depreciation & 3 Years, Third Party]	Extended Warranty Cost [additional 2 Years or 0 Kms]	Maintenance Package Cost [2 Years & 20,000 Kms]	On Road Cost		
Hycross Hybrid	2L ZX (O)	INNHA	2897000	28970	578821	5500	7500	143642	42924	NYD	3,704,357		
	2L ZX (O)	INNHA Pearl White	2906500	29065	580719	5500	7500	144003	42924	NYD	3,716,211		
	2L ZX	INNHB	2833000	28330	566034	5500	7500	141211	42924	NYD	3,624,499		
	2L ZX	INNHB Pearl White	2842500	28425	567932	5500	7500	141572	42924	NYD	3,636,353		
	2L VX 7S	INNHE	2401000	24010	479720	5500	7500	124799	42924	NYD	3,085,453		
	2L VX 7S	INNHE Pearl White	2410500	24105	481618	5500	7500	125160	42924	NYD	3,097,307		
	2L VX 8S	INNHF	2406000	24060	480719	5500	7500	125343	42924	NYD	3,092,046		
	2L VX 8S	INNHF Pearl White	2415500	24155	482617	5500	7500	125704	42924	NYD	3,103,900		
Hycross Gasolin	2L GX 7S	INNHI	1915000	19150	361361	5500	7500	104219	29091	NYD	2,441,821		
	2L GX 7S	INNHI Pearl White	1924500	19245	363154	5500	7500	104558	29091	NYD	2,453,548		
	2L GX 8S	INNHI	1920000	19200	362304	5500	7500	104751	29091	NYD	2,448,346		
	2L GX 8S	INNHI Pearl White	1929500	19295	364097	5500	7500	105091	29091	NYD	2,460,074		
	2L G SLF 7S	INNDR	1830000	18300	345321	5500	7500	101180	29091	NYD	2,336,892		
	2L G SLF 7S	INNDR Pearl White	1839500	18395	347114	5500	7500	101520	29091	NYD	2,348,820		
	2L G SLF 8S	INNDR	1835000	18350	346265	5500	7500	101713	29091	NYD	2,343,419		
	2L G SLF 8S	INNDR Pearl White	1844500	18445	348058	5500	7500	102052	29091	NYD	2,355,146		
Including Hypothecation, HSRP, FasTag, Safety Cess & Service Charges *Mud Flap, Floor Mat													
Terms & Conditions													
1. Toyota Kirloskar Motor reserves right to change, model, color, specification etc. without prior notice													
2. Prices quoted are indicative only and price prevailing at the time of delivery will be applicable													
3. Minimum Booking Amount: Rs. 50,000													
4. *Tax Collected at Source of 1% will be collected, if Ex-Showroom Price cross Rs. 1,000,000													
5. Add on features available on Insurance Policy													
6. Variation of Insurance premium based on Insurance provider, Consumers are free to avail insurance services from any insurance company													
7. Road Tax to be calculated on ex-showroom cost													
8. Cancellation Policy, Amount will be refunded in 15 days from Date of Cancellation													
9. RTGS / Payment / Cheque / Demand Draft Details of Dealer: M/s. MOTOR WORLD PRIVATE LTD													
10. Cash collection of upto Rs.2 Lakhs only. Cash should be deposited at cash counter of our showroom only and collect original receipt.													
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								Follow Us		For Directions			
													
								Visit Our Website					

Exterior Color Options

The Hycross is offered in **seven** main colors:

1.Blackish Ageha Glass Flake: A signature dark blue-black that shifts in sunlight.

2.Platinum White Pearl: A premium metallic white.

3.Sparkling Black Pearl Crystal Shine: A deep black with a pearl finish.

4.Attitude Black Mica: A solid, glossy black.

5.Silver Metallic: A standard bright silver.

6.Avant Garde Bronze Metallic: A sophisticated brownish-grey.

7.Super White: A solid non-metallic white, often favored for fleet use.

Interior Schemes

•**Black Fabric:** Standard on base G and GX trims.

•**Chestnut & Black (Dual Tone):** Available on mid-range GX (O), VX, and VX (O) trims.

•**Dark Chestnut Art Leather:** Exclusive to the high-end ZX and ZX (O) trims.



Pricing

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Trim / Variant	Powertrain	Drivetrain / Notes	MSRP (USD)
Ranger XLT 4x2 (SuperCrew 5' box)	2.3L EcoBoost I4 - 270 hp / 310 lb-ft [cars.usnews.com] , [fordauthority.com]	RWD (4x2), 10-speed automatic [cars.com]	\$36,075 [cars.com]
Ranger Raptor	3.0L twin-turbo EcoBoost V6 - 405 hp / 430 lb-ft [autoblog.com] , [carbuzz.com]	Standard 4WD, off-road performance suspension [autoblog.com]	\$57,070 [cars.com] , [ford.com]

Trim / Variant	Powertrain	Drivetrain / Notes	MSRP (VND)
Ranger XLS 4x2 AT	2.0L Turbo Diesel I4 (Single Turbo TDCi) - ~170 PS / 405 Nm [ford.com.vn] , [fordbinhtrieu.com.vn]	RWD (4x2), 6AT or 6MT [ford.com.vn]	VND 707,000,000 (~\$28,280)
Ranger Raptor (Diesel - legacy)	2.0L Bi-Turbo Diesel I4 - ~210 PS / 500 Nm [xe.today]	4WD, 10AT, performance suspension [oto.com.vn]	VND 1,299,000,000 (~\$51,960)
Ranger Raptor (NEW Gasoline 2026)	3.0L V6 EcoBoost Twin-Turbo Gasoline - ~397 PS / 583 Nm [xe.today] , [vnauto.net]	Full-time 4WD, 10AT, FOX adaptive suspension [vnauto.net]	VND 1,448,000,000 (~\$57,920)



AFS; Serviceability; Maintenance

Nice sources to consider:

- US Market: The Car Care Nut Reviews - YouTube

Conclusion





Kết luận – Key Message

- 3 điểm mạnh nhất
- Tính định vị
- Lời khẳng định cuối (closing statement)



Toyota Tacoma 4th Gen (N400; 2023)

Top 3 Strengths

- **Exceptional reputation for durability & reliability** → proven body-on-frame platform and long-term ownership trust
- **Strong off-road capability across trims** → supported by Crawl Control, Multi-Terrain Select, and TRD hardware
- **Wide product coverage (SR → TRD Pro)** → one model serving fleet, lifestyle, and premium customers

Positioning

- **Segment:** Midsize pickup truck
- **Core Identity:** “Durability-first, capability-driven truck”
- **Brand Positioning:** Benchmark for **resale value, reliability, and off-road credibility**
- **Market Role:** Bridges **daily usability + adventure capability + light-dutywork**

Closing Statement

- Tacoma represents a **highly engineered, modular truck platform** designed to support **multiple lifestyles with one product**
- It reinforces Toyota’s philosophy of **trust, durability, and authentic capability over complexity**
- → A **market benchmark in reliability and off-road performance**, but with opportunity gaps in **electrification and digital differentiation**

Takeaway

- Tacoma = “**Capability + Reliability Benchmark**” in midsize pickup segment
- Future competitors must combine this **DNA with next-gen electrification & smart features**

Thank You

